Terrain Classification for Small Legged Robots

Team B-29 (2 members) | B.Tech CSE (AI & ML) | Machine Learning mini-project
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1. Problem Statement

A legged robot needs to recognize the surface under its feet to move reliably. This project classifies the terrain encountered by a small legged robot into one of eight categories (asphalt, cork, grass, gravel, lab, laminate wood, pebble and sand) from tactile sensor data. Three classical machine-learning models are compared against a convolutional neural network (CNN).

2. Dataset

Samples	5,990	**Training / validation / test**	4,193 / 898 / 899 (70 / 15 / 15, stratified)
Features	447 per sample	**Tactile readings**	414 values (6 channels × 69 readings)
Classes	8 terrains	**Engineered features**	33

The notebook asserts the 5,990 × 447 shape, checks that no values are missing or non-finite, and verifies that the two label sources match. The original source of the dataset is not documented in the repository.

3. Methodology

Data are split with stratified sampling (fixed seed 42). Normalization parameters are estimated from the training data only and applied unchanged to validation and test data. The validation set is used for model selection; the held-out test set is used once for final evaluation. Models are compared using accuracy and macro F1-score.

4. Implementation

Experiments were run in Python (scikit-learn, PyTorch, pandas, NumPy). The four models are:

- **Logistic Regression:** standardized 447 features, max 2,000 iterations.
- **SVM (RBF):** standardized 447 features, C = 10, gamma = “scale”.
- **Random Forest:** all 447 features, 400 trees, minimum leaf size 2, balanced class weights.
- **CNN:** four 1D-style convolution layers (kernel 1×5, 24 filters each) with max-pooling and dropout 0.5 over the 6 × 69 tactile signal; the flattened output is concatenated with all 33 engineered features, then a 50-unit dense layer and an 8-class output. Trained with Adam (learning rate 1e-4, batch size 64, up to 100 epochs) with early stopping on validation accuracy (patience 12); the best-validation weights are restored before testing.

5. Results

Model	Accuracy	Macro F1
Random Forest	**77.09%**	**0.7698**
SVM (RBF)	70.41%	0.7093
CNN	61.96%	0.5900
Logistic Regression	56.06%	0.5593

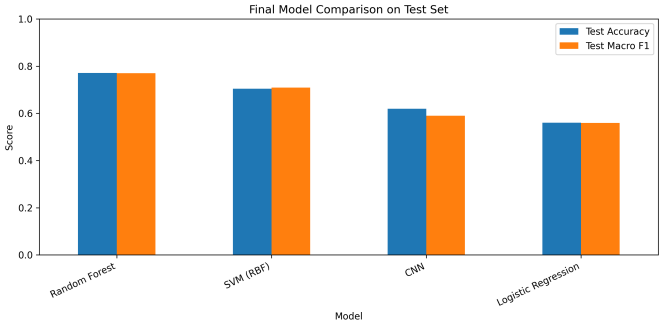


Table 1 and Figure 1: final test results on the 899 held-out samples (single stratified split).

Random Forest achieved the highest test accuracy (77.09%) and macro F1 (0.7698), about 15.13 percentage points above the CNN. Figure 2 shows the CNN's test confusion matrix: recall is strongest for sand (98%) and grass (83%), then laminate wood (74%), pebble (64%), gravel (59%), lab (58%) and asphalt (49%), and weakest for cork (11%), which is most often predicted as grass, gravel or pebble.

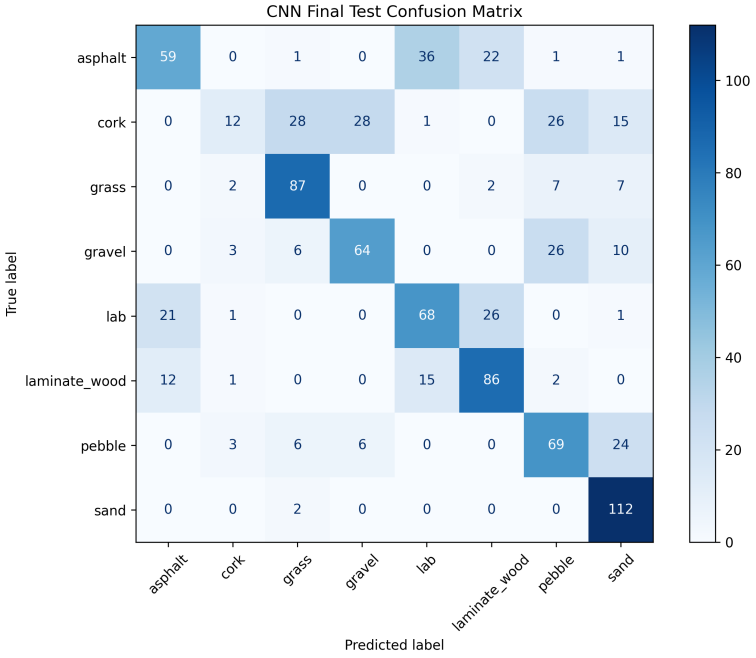


Figure 2: CNN final test confusion matrix.

6. Limitations

- Results come from one stratified split and one training run, so they may vary with other splits or random seeds.
- The CNN's low cork recall is a key weakness. No cross-validation, feature-importance analysis or real-robot testing was performed.
- The comparison holds for this dataset, preprocessing and model configuration; it does not show that Random Forest is better than CNNs in general.

7. Conclusion

Random Forest performed best of the four models, reaching 77.09% held-out test accuracy and 0.7698 macro F1, ahead of SVM (70.41%), the CNN (61.96%) and Logistic Regression (56.06%). The CNN was strong on sand and grass but weak on cork. The results support benchmarking deep learning against classical baselines rather than assuming a neural network will win. Future work: repeated splits or cross-validation, feature importance, alternative architectures, and class-specific error analysis.